State or Market? Survey of preferences for financing biological diversity in SE (#273748)

Author(s)
Jens Christiansen (Lund University Centre for Sustainability Studies) - jens.christiansen@lucsus.lu.se
Nils Droste (Copenhagen University) - nd@ifro.ku.dk
Yohei Mitani (Kyoto University) - mitani.yohei.7w@kyoto-u.ac.jp

Pre-registered on:
2026/02/15 23:21 (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

How do financing designs influence Swedish citizens' willingness to fund biodiversity restoration and conservation?

3) Describe the key dependent variable(s) specifying how they will be measured.

The main part of this study is a discrete choice experiment. Respondents will be presented with two policy options for funding biodiversity at a time. The respondent will choose a policy option or a status quo option. The dependent variable is a series of discrete choices of the most preferred option among three options shown in each choice set. The status quo option is constant over eight choice sets, whereas two policy options, characterized by four attributes, varied across choice sets.

The attributes for the policy options are:
Type of payment: An increase in your net tax contribution, with an explicit purpose of supporting biodiversity conservation and restoration; voluntary private donation to civil society organizations that work with restoring and conserving biodiversity; a higher price for a certified commodity, which is produced in a biodiversity friendly way; state requires companies to offset their biodiversity impact and market prices would increase as a consequence.
Ownership of protected area: Biodiversity protection on publicly owned land through creating protected areas; biodiversity protection on large scale industrial land by paying for setting aside land from production; biodiversity protection on small-scale land owners' land by compensating for losses in income
Monitoring and reporting: Public monitoring and enforcement by a state authority or agency; monitoring by the private landowner themself and a third-party auditor, such as e.g. a consultancy
Amount: An amount of yearly payments at four levels, 492SEK, 2460SEK, 4920SEK, and 7380SEK

Respondents will be presented with a total of 8 pairs of policy options.

4) How many and which conditions will participants be assigned to?

For the choice sets, we use a 4 x 3 x 2 x 4 full factorial design. This results in 96 choice sets, which are grouped into 12 groups. Each participant will consequently be presented with 8 choice sets. The levels are defined in section 3 above. In addition, participants will be presented with 8 likert scale questions on a scale from 1 to 5.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will employ choice modelling on the discrete experiment on their willingness to pay for different biodiversity financing options, including conditional logit model and mixed logit model specifications including socio-demographics such as available income.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Our survey design makes it impossible for respondents to produce outlier observations. We have no specific rules on excluding outliers but cannot estimate willingness to pay for those that opt out on all choice sets.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We aim for a representative sample of the Swedish population by recruiting around 2000 respondents via an online survey.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will consider effect heterogeneity regarding socio-economic background variables, consider effects of income, gender, geographic location, values etc.